

Ayon Ganguly

Associate Professor
Department of Mathematics
Indian Institute of Technology Guwahati



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Research Interests

Lifetime data analysis
Truncated data
Cure rate models

Censoring schemes
Competing risks
Estimation of area under ROC

Step-stress life tests
Load sharing systems

Employment

Associate Professor

Department of Mathematics
Indian Institute of Technology Guwahati
India

October, 2026 – present

Assistant Professor

Department of Mathematics
Indian Institute of Technology Guwahati
India

June, 2016 – October, 2026

Assistant Professor (on contract)

Department of Applied Sciences
Indian Institute of Information Technology Allahabad
India

January, 2026 – June, 2016

Visiting Faculty (Adhoc)

Department of Applied Sciences
Indian Institute of Information Technology Allahabad
India

December, 2014 – January, 2016

Post-doctoral fellow (Dr. DS Kothari Fellowship, UGC)

Department of Statistics
Savitribai Phule Pune University, Pune
India

September, 2013 – December, 2014

Visiting Scientist

Indian Statistical Institute, Chennai
India

April, 2013 – September, 2013

Education

PhD in Statistics

Department of Mathematics and Statistics
Indian Institute of Technology Kanpur
India

2013

Master of Science in Statistics

Department of Mathematics and Statistics
Indian Institute of Technology Kanpur
India

2008

Bachelor of Science in Statistics

Department of Statistics
Maulana Azad College, Kolkata
India

2006

Publications

Total Publications: 16

- Ganguly, A., Sultana, F., Kundu, D., & Pal, A. (2026). A model based on mixture of weibull distributions for depending competing risks data in the presence of long-term survivors, and its application to malignant melanoma cancer data. *Statistics in Medicine*, 45(6-7), e70466. <https://doi.org/10.1002/sim.70466>
- Agrawal, A., Ganguly, A., & Mitra, D. (2025). Modelling competing risks data by a bivariate model with singularity originating from a shock model using the lehmann family of distributions. *Research in Statistics*, 3(1), 2480181. <https://doi.org/10.1080/27684520.2025.2480181>
- Agrawal, A., Mitra, D., & Ganguly, A. (2025). A model for failure-time data with two dependent failure modes and prediction of future failures. *Quality Engineering*, 37(1), 1–14. <https://doi.org/10.1080/08982112.2024.2321839>
- Biswas, S., Ganguly, A., & Mitra, D. (2025). Reliability analysis of load-sharing systems using a flexible model with piecewise linear functions. *Applied Stochastic Models in Business and Industry*, 41(1), e2934. <https://doi.org/10.1002/asmb.2934>
- Ganguly, A., Mitra, D., Balakrishnan, N., & Kundu, D. (2024). A flexible model based on piecewise linear approximation for the analysis of left truncated right censored data with covariates, and applications to worcester heart attack study data and channing house data. *Statistics in Medicine*, 43, 233–255. <https://doi.org/10.1002/sim.9954>
- Samanta, D., Ganguly, A., Gupta, A., & Kundu, D. (2019). On classical and Bayesian order restricted inference for multiple exponential step stress model. *Statistics*, 53(1), 177–195. <https://doi.org/10.1080/02331888.2018.1526285>
- Samanta, D., Kundu, D., & Ganguly, A. (2018). Order restricted Bayesian analysis of a simple step stress model. *Sankhya B*, 80, 195–221. <https://doi.org/10.1007/s13571-017-0139-9>
- Hanagal, D. D., Pandey, A., & Ganguly, A. (2017). Correlated gamma frailty models for bivariate survival data. *Communications in Statistics - Simulation and Computation*, 46(5), 3627–3644. <https://doi.org/10.1080/03610918.2015.1085559>
- Koley, A., Kundu, D., & Ganguly, A. (2017). Analysis of type-II hybrid censored competing risks data. *Statistics*, 51(6), 1304–1325. <https://doi.org/10.1080/02331888.2017.1358271>

- Kundu, D., Mitra, D., & Ganguly, A. (2017). Analysis of left truncated and right censored competing risks data. *Computational Statistics & Data Analysis*, 108, 12–26. <https://doi.org/10.1016/j.csda.2016.10.020>
- Samanta, D., Ganguly, A., Kundu, D., & Mitra, S. (2017). Order restricted bayesian inference for exponential simple step-stress model. *Communications in Statistics - Simulation and Computation*, 46(2), 1113–1135. <https://doi.org/10.1080/03610918.2014.992540>
- Ganguly, A., & Kundu, D. (2016). Analysis of simple step-stress model in presence of competing risks. *Journal of Statistical Computation and Simulation*, 86(10), 1989–2006. <https://doi.org/10.1080/00949655.2015.1096362>
- Ganguly, A., Kundu, D., & Mitra, S. (2015). Bayesian analysis of a simple step-stress model under weibull lifetimes. *IEEE Transactions on Reliability*, 64(1), 473–485. <https://doi.org/10.1109/TR.2014.2368871>
- Kundu, D., Samanta, D., Ganguly, A., & Mitra, S. (2013). Bayesian analysis of different hybrid and progressive life tests. *Communications in Statistics - Simulation and Computation*, 42(9), 2160–2173. <https://doi.org/10.1080/03610918.2011.654027>
- Mitra, S., Ganguly, A., Samanta, D., & Kundu, D. (2013). On the simple step-stress model for two-parameter exponential distribution. *Statistical Methodology*, 15, 95–114. <https://doi.org/10.1016/j.stamet.2013.04.003>
- Ganguly, A., Mitra, S., Samanta, D., & Kundu, D. (2012). Exact inference for the two-parameter exponential distribution under type-II hybrid censoring. *Journal of Statistical Planning and Inference*, 142(3), 613–625. <https://doi.org/10.1016/j.jspi.2011.08.001>

Submitted Articles Total submitted articles: 4

- Biswas, S., Ganguly, A., & Mitra, D. (2026). A doubly-flexible model based on generalized gamma frailty for two-component load-sharing systems.
- Biswas, S., Ganguly, A., & Mitra, D. (2026). Bayesian analysis of generalized freund's bivariate model for a two-component load-sharing system.
- Saha, S., Mitra, D., & Ganguly, A. (2026). Analysis of bivariate left truncated right censored data using flexible families of distributions and its applications to the canadian insurance data.
- Sanyal, A., Kattumannil, S. K., & Ganguly, A. (2026). Time-dependent two-way partial AUC and partial youden index estimator for right censored data.

Book Published

- Kundu, D., & Ganguly, A. (2017). *Analysis of step-stress models: Existing results and recent developments*. Elsevier/Academic Press.

Doctoral Students Total doctoral students: 3

Dr. Shilpi Biswas

Thesis title: Statistical analysis of load-sharing systems

Role: Supervisor

Current Position: Independent researcher

2019 – 2025

Mr. Sourav Saha

Area: Statistical analysis of bivariate left truncated right censored data

Role: Co-supervisor (jointly with Prof. Debanjan Mitra, IIM Udaipur)

Current Position: Doctoral Student, IIT Guwahati

2022 – present

Mr. Arpan Sanyal

Area: AUC estimation for a ROC

Role: Co-supervisor (jointly with Dr. S. K. Kattumannil, ISI Channai)

Current Position: Doctoral Student, IIT Guwahati

2023 – present

Research Grant

Mathematical Research Impact Centric Support (MATRICS)

Title: Investigation of Statistical Aspects of Step-stress Life Testing

Funding agency: Science and Engineering Research Board, DST, Government of India

Amount: Rs. 6,60,000.00

2018 – 2021

Editorial Activity

Mathematical Methods of Statistics, Springer Nature

Role: Associate Editor

2025 – 2026

Computer Proficiency

Latex, R, C++

Language Proficiency

Bangla (Native), English, Hindi

References

Will be provided upon request.